KERALA DIPLOMA CURRICULUM REVISION 2026

Course 4111 | 4 Credits: 4 | Program Core | Source-grounded

Fluid Mechanics & Heat Transfer

 To establish fundamental knowledge of basic fluid mechanics and address specific

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4111 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4111.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Tool & Die Engineering
Course code	4111
Course title	Fluid Mechanics & Heat Transfer
Semester	4 Credits: 4
Course category	Program Core

Item	Official information
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To establish fundamental knowledge of basic fluid mechanics and address specific
- To familiarize students with the relevance of fluid dynamics to many engineering
- To appreciate basics of hydraulic and pneumatic systems
- To introduce the basics of thermodynamics and modes of heat transfer

Course outcomes

CO	Official outcome	Study level
CO1	15 Applying devices and hydrostatic forces Investigate the applications of laws of conservation of momentum, mass and energy in fluid flow	See official syllabus
CO2	14 Applying systems and the characteristics of flow though pipes, orifices and notches Distinguish the components and basic circuits in	See official syllabus
CO3	15 Understanding hydraulic and pneumatic systems Illustrate the basic thermodynamics principles and	See official syllabus
CO4	14 Applying different modes of heat transfer	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2
CO2	CO2 3 3
CO3	CO3 2
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	forces	4	As specified
Module 2	and energy in fluid flow systems and the characteristics of flow though	4	As specified
Module 3	systems	5	As specified
Module 4	heat transfer	3	As specified

MODULE 1

forces

This module is presented from the official syllabus scope for Fluid Mechanics & Heat Transfer. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: forces
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

Explain fluid properties 3 Understanding Explain the Fluid Pressure and the methods of

Explain fluid properties 3 Understanding Explain the Fluid Pressure and the methods of is an official topic in Module 1 of Fluid Mechanics & Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain fluid properties 3 Understanding Explain the Fluid Pressure and the methods of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain fluid properties 3 understanding explain the fluid pressure and the methods of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain fluid properties 3 Understanding Explain the Fluid Pressure and the methods of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain fluid properties 3 Understanding Explain the Fluid Pressure and the methods of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Understanding measurement Illustrate the principle of working of various

4 Understanding measurement Illustrate the principle of working of various is an official topic in Module 1 of Fluid Mechanics & Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding measurement Illustrate the principle of working of various using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding measurement illustrate the principle of working of various should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding measurement Illustrate the principle of working of various means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-[illegible] 4 Understanding measurement Illustrate the principle of working of various [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

pressure measuring devices and solve simple 4 Applying problems on those Interpret the terms total pressure & buoyancy

pressure measuring devices and solve simple 4 Applying problems on those Interpret the terms total pressure & buoyancy is an official topic in Module 1 of Fluid Mechanics & Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify pressure measuring devices and solve simple 4 Applying problems on those Interpret the terms total pressure & buoyancy using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, pressure measuring devices and solve simple 4 applying problems on those interpret the terms total pressure & buoyancy should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What pressure measuring devices and solve simple 4 Applying problems on those Interpret the terms total pressure & buoyancy means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

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and calculate the total pressure on immersed 4 Applying surfaces Properties of fluid: Density, Specific gravity, Specific Weight, Specific Volume, Dynamic Viscosity, Kinematic Viscosity, Surface tension, Capillarity, Vapour Pressure, Compressibility. Numerical problems on fluid properties. Fluid Pressure & Pressure Measurement: Fluid pressure, Pressure head, Pressure intensity, Concept of vacuum and gauge pressures, atmospheric pressure, absolute pressure, Simple and differential manometers, Bourdons pressure gauge, Concept of Total pressure on immersed bodies, center of pressure, Simple problems on pressure measurements Buoyancy and flotation, Archimedes' principle, stability of immersed and floating bodies, determination of metacentric height. Problems on total pressure. Investigate the applications of laws of conservation of momentum, mass

and calculate the total pressure on immersed 4 Applying surfaces Properties of fluid: Density, Specific gravity, Specific Weight, Specific Volume, Dynamic Viscosity, Kinematic Viscosity, Surface tension, Capillarity, Vapour Pressure, Compressibility. Numerical problems on fluid properties. Fluid Pressure & Pressure Measurement: Fluid pressure, Pressure head, Pressure intensity, Concept of vacuum and gauge pressures, atmospheric pressure, absolute pressure, Simple and differential manometers, Bourdons pressure gauge, Concept of Total pressure on immersed bodies, center of pressure, Simple problems on pressure measurements Buoyancy and flotation, Archimedes' principle, stability of immersed and floating bodies, determination of metacentric height. Problems on total pressure. Investigate the applications of laws of conservation of momentum, mass is an official topic in Module 1 of Fluid Mechanics & Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify and calculate the total pressure on immersed 4 Applying surfaces Properties of fluid: Density, Specific gravity, Specific Weight, Specific Volume, Dynamic Viscosity, Kinematic Viscosity, Surface tension, Capillarity, Vapour Pressure, Compressibility. Numerical problems on fluid properties. Fluid Pressure & Pressure Measurement: Fluid pressure, Pressure head, Pressure intensity, Concept of vacuum and gauge pressures, atmospheric pressure,

absolute pressure, Simple and differential manometers, Bourdons pressure gauge, Concept of Total pressure on immersed bodies, center of pressure, Simple problems on pressure measurements Buoyancy and flotation, Archimedes' principle, stability of immersed and floating bodies, determination of metacentric height. Problems on total pressure. Investigate the applications of laws of conservation of momentum, mass using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, and calculate the total pressure on immersed 4 applying surfaces properties of fluid: density, specific gravity, specific weight, specific volume, dynamic viscosity, kinematic viscosity, surface tension, capillarity, vapour pressure, compressibility. numerical problems on fluid properties. fluid pressure & pressure measurement: fluid pressure, pressure head, pressure intensity, concept of vacuum and gauge pressures, atmospheric pressure, absolute pressure, simple and differential manometers, bourdons pressure gauge, concept of total pressure on immersed bodies, center of pressure, simple problems on pressure measurements buoyancy and flotation, archimedes' principle, stability of immersed and floating bodies, determination of metacentric height. problems on total pressure. investigate the applications of laws of conservation of momentum, mass should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What and calculate the total pressure on immersed 4 Applying surfaces Properties of fluid: Density, Specific gravity, Specific Weight, Specific Volume, Dynamic Viscosity, Kinematic Viscosity, Surface tension, Capillarity, Vapour Pressure, Compressibility. Numerical problems on fluid properties. Fluid Pressure & Pressure Measurement: Fluid pressure, Pressure head, Pressure intensity, Concept of vacuum and gauge pressures, atmospheric pressure, absolute pressure, Simple and differential manometers, Bourdons pressure gauge, Concept of Total pressure on immersed bodies, center of pressure, Simple problems on pressure measurements Buoyancy and flotation, Archimedes' principle, stability of immersed and floating bodies, determination of metacentric height. Problems on total pressure. Investigate the applications of laws of conservation of momentum, mass means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ and calculate the total pressure on immersed 4 Applying surfaces Properties of fluid: Density, Specific gravity, Specific Weight, Specific Volume, Dynamic Viscosity, Kinematic Viscosity, Surface tension, Capillarity, Vapour Pressure, Compressibility. Numerical problems on fluid properties. Fluid Pressure & Pressure Measurement: Fluid pressure, Pressure head, Pressure intensity, Concept of vacuum and gauge pressures, atmospheric pressure, absolute pressure, Simple and differential manometers, Bourdons pressure gauge, Concept of Total pressure on immersed bodies, center of pressure, Simple problems on pressure measurements Buoyancy and flotation, Archimedes’ principle, stability of immersed and floating bodies, determination of metacentric height. Problems on total pressure. Investigate the applications of laws of conservation of momentum, mass □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

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forces

  M1.01     Explain fluid properties                             3
Understanding

            Explain the Fluid Pressure and the methods of
  M1.02                                                          4
Understanding
            measurement
            Illustrate the principle of working of various
  M1.03     pressure measuring devices and solve simple          4
Applying
            problems on those
            Interpret the terms total pressure & buoyancy
  M1.04     and calculate the total pressure on immersed         4
Applying
            surfaces

Contents:
Properties of fluid: Density, Specific gravity, Specific Weight, Specific Volume,
Dynamic
Viscosity, Kinematic Viscosity, Surface tension, Capillarity, Vapour Pressure,
Compressibility. Numerical problems on fluid properties.
Fluid Pressure & Pressure Measurement: Fluid pressure, Pressure head, Pressure
intensity,
Concept of vacuum and gauge pressures, atmospheric pressure, absolute pressure,
Simple
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

and energy in fluid flow systems and the characteristics of flow though

This module is presented from the official syllabus scope for Fluid Mechanics & Heat Transfer. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: and energy in fluid flow systems and the characteristics of flow though
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Describe the types of fluid flows 2 Understanding Apply the momentum and energy equations to

Describe the types of fluid flows 2 Understanding Apply the momentum and energy equations to is an official topic in Module 2 of Fluid Mechanics & Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the types of fluid flows 2 Understanding Apply the momentum and energy equations to using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the types of fluid flows 2 understanding apply the momentum and energy equations to should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the types of fluid flows 2 Understanding Apply the momentum and energy equations to means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Describe the types of fluid flows 2 Understanding Apply the momentum and energy equations to □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

fluid flow problems based on an analysis of 4 Applying the various system specifications Illustrate the concept of flow through pipes,

fluid flow problems based on an analysis of 4 Applying the various system specifications Illustrate the concept of flow through pipes, is an official topic in Module 2 of Fluid Mechanics & Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify fluid flow problems based on an analysis of 4 Applying the various system specifications Illustrate the concept of flow through pipes, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, fluid flow problems based on an analysis of 4 applying the various system specifications illustrate the concept of flow through pipes, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What fluid flow problems based on an analysis of 4 Applying the various system specifications Illustrate the concept of flow through pipes, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] fluid flow problems based on an analysis of 4 Applying the various system specifications Illustrate the concept of flow through pipes, [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

4 Applying orifices & notches List the losses of head in pipes and identify

4 Applying orifices & notches List the losses of head in pipes and identify is an official topic in Module 2 of Fluid Mechanics & Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying orifices & notches List the losses of head in pipes and identify using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying orifices & notches list the losses of head in pipes and identify should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying orifices & notches List the losses of head in pipes and identify means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-□□ 4 Applying orifices & notches List the losses of head in pipes and identify □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Major losses and Minor losses and solve 4 Applying problems Fluid Flow: Types of fluid flows, Path line and Stream line, Continuity equation, Bernoulli's theorem, Principle of operation of Venturimeter, Orifice meter and Pitot tube, Derivations for discharge, coefficient of discharge, numerical problems. Flow Through Pipes, Orifices & Notches: Darcy's equation and Chezy's equation for frictional losses (No derivation), Minor losses in pipes, Numerical problems to estimate major and minor losses. Orifices: types of orifices, Vena contracta, coefficient of contraction, coefficient of velocity, coefficient of discharge, simple problems. Notches: Types –Rectangular notches , triangular notch , trapezoidal notch , discharge over notches – equations only, simple numerical problems Distinguish the components and basic circuits in hydraulic and pneumatic

Major losses and Minor losses and solve 4 Applying problems Fluid Flow: Types of fluid flows, Path line and Stream line, Continuity equation, Bernoulli's theorem, Principle of operation of Venturimeter, Orifice meter and Pitot tube, Derivations for discharge, coefficient of discharge, numerical problems. Flow Through Pipes, Orifices & Notches: Darcy's equation and Chezy's equation for frictional losses (No derivation), Minor losses in pipes, Numerical problems to estimate major and minor losses. Orifices: types of orifices, Vena contracta, coefficient of contraction, coefficient of velocity, coefficient of discharge, simple problems. Notches: Types – Rectangular notches , triangular notch , trapezoidal notch , discharge over notches – equations only, simple numerical problems Distinguish the components and basic circuits in hydraulic and pneumatic is an official topic in Module 2 of Fluid Mechanics & Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Major losses and Minor losses and solve 4 Applying problems Fluid Flow: Types of fluid flows, Path line and Stream line, Continuity equation, Bernoulli's theorem, Principle of operation of Venturimeter, Orifice meter and Pitot tube, Derivations for discharge, coefficient of discharge, numerical problems. Flow Through Pipes, Orifices & Notches: Darcy's equation and Chezy's equation for frictional losses (No derivation), Minor losses in pipes, Numerical problems to estimate major and minor losses. Orifices: types of orifices, Vena contracta, coefficient of contraction, coefficient of velocity, coefficient of discharge, simple problems. Notches: Types –Rectangular notches , triangular notch , trapezoidal notch , discharge over notches – equations only, simple numerical problems Distinguish the components and basic circuits in hydraulic and pneumatic using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, major losses and minor losses and solve 4 applying problems fluid flow: types of fluid flows, path line and stream line, continuity equation, bernoulli’s theorem, principle of operation of venturimeter, orifice meter and pitot tube, derivations for discharge, coefficient of discharge, numerical problems. flow through pipes, orifices & notches: darcy’s equation and chezy’s equation for frictional losses (no derivation), minor losses in pipes, numerical problems to estimate major and minor losses. orifices: types of orifices, vena contracta, coefficient of contraction, coefficient of velocity, coefficient of discharge, simple problems. notches: types –rectangular notches , triangular notch , trapezoidal notch , discharge over notches – equations only, simple numerical problems distinguish the components and basic circuits in hydraulic and pneumatic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Major losses and Minor losses and solve 4 Applying problems Fluid Flow: Types of fluid flows, Path line and Stream line, Continuity equation, Bernoulli’s theorem, Principle of operation of Venturimeter, Orifice meter and Pitot tube, Derivations for discharge, coefficient of discharge, numerical problems. Flow Through Pipes, Orifices & Notches: Darcy’s equation and Chezy’s equation for frictional losses (No derivation), Minor losses in pipes, Numerical problems to estimate major and minor losses. Orifices: types of orifices, Vena contracta, coefficient of contraction, coefficient of velocity, coefficient of discharge, simple problems. Notches: Types –Rectangular notches , triangular notch , trapezoidal notch , discharge over notches – equations only, simple numerical problems Distinguish the components and basic circuits in hydraulic and pneumatic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Major losses and Minor losses and solve 4 Applying problems Fluid Flow: Types of fluid flows, Path line and Stream line, Continuity equation, Bernoulli’s theorem, Principle of operation of Venturimeter, Orifice meter and Pitot tube, Derivations for discharge, coefficient of discharge, numerical problems. Flow Through Pipes, Orifices & Notches: Darcy’s equation and Chezy’s equation for frictional losses (No derivation), Minor losses in pipes, Numerical problems to estimate major and minor losses. Orifices: types of orifices, Vena contracta, coefficient of contraction, coefficient of velocity, coefficient of discharge, simple problems. Notches: Types –Rectangular notches , triangular notch , trapezoidal notch , discharge over notches – equations only, simple numerical problems Distinguish the components and basic circuits in hydraulic and pneumatic □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
and energy in fluid flow systems and the characteristics of flow though
            pipes, orifices and notches
  M2.01     Describe the types of fluid flows                   2
Understanding
            Apply the momentum and energy equations to
  M2.02     fluid flow problems based on an analysis of           4
Applying
            the various system specifications
            Illustrate the concept of flow through pipes,
  M2.03                                                           4
Applying
            orifices & notches
            List the losses of head in pipes and identify
  M2.04     Major losses and Minor losses and solve               4
Applying
            problems
            Series Test I                                         1

Contents:
Fluid Flow: Types of fluid flows, Path line and Stream line, Continuity equation,
Bernoulli's theorem, Principle of operation of Venturimeter, Orifice meter and
Pitot tube,
Derivations for discharge, coefficient of discharge, numerical problems.
Flow Through Pipes, Orifices & Notches: Darcy's equation and Chezy's equation for
frictional losses (No derivation), Minor losses in pipes, Numerical problems to
estimate
major and minor losses. Orifices: types of orifices, Vena contracta, coefficient
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

systems

This module is presented from the official syllabus scope for Fluid Mechanics & Heat Transfer. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: systems
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

1 Understanding and its applications Summarize the features and functions of

1 Understanding and its applications Summarize the features and functions of is an official topic in Module 3 of Fluid Mechanics & Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding and its applications Summarize the features and functions of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding and its applications summarize the features and functions of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding and its applications Summarize the features and functions of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-□□ 1 Understanding and its applications Summarize the features and functions of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

various components of hydraulic and 4 Understanding pneumatic systems Identify various Hydraulic and Pneumatic

various components of hydraulic and 4 Understanding pneumatic systems Identify various Hydraulic and Pneumatic is an official topic in Module 3 of Fluid Mechanics & Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify various components of hydraulic and 4 Understanding pneumatic systems Identify various Hydraulic and Pneumatic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, various components of hydraulic and 4 understanding pneumatic systems identify various hydraulic and pneumatic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What various components of hydraulic and 4 Understanding pneumatic systems Identify various Hydraulic and Pneumatic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ various components of hydraulic and 4 Understanding pneumatic systems Identify various Hydraulic and Pneumatic □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

2 Remembering symbols Describe the types of valves used in hydraulic

2 Remembering symbols Describe the types of valves used in hydraulic is an official topic in Module 3 of Fluid Mechanics & Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Remembering symbols Describe the types of valves used in hydraulic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 remembering symbols describe the types of valves used in hydraulic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Remembering symbols Describe the types of valves used in hydraulic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-□□ 2 Remembering symbols Describe the types of valves used in hydraulic □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

4 Understanding and pneumatic systems Explain the working of different types of

4 Understanding and pneumatic systems Explain the working of different types of is an official topic in Module 3 of Fluid Mechanics & Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding and pneumatic systems Explain the working of different types of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding and pneumatic systems explain the working of different types of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding and pneumatic systems Explain the working of different types of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 4 Understanding and pneumatic systems Explain the working of different types of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

4 Understanding hydraulic and pneumatic circuits and systems Hydraulics: Introduction, basic concepts, Hydraulic fluids, Classification and properties of hydraulic fluids, Basic components, Fluid power generators, i.e. Gear, Vane, Piston pumps, linear and Rotary Actuators, Direction Control Valves, types, actuation methods, pressure control valves; pressure reducing valves, pressure relief valve, Flow control valves - simple and pressure compensated type. Circuit symbols, automatic cylinder reciprocating circuit & hydraulic circuit for robotic arm. Pneumatics: Introduction, Basic components, Source, storage and distribution, treatment of compressed air, linear and Rotary actuators, Direction control valves – types, actuation methods, pressure control valves, logic devices – twin pressure valve, shutter valve, time delay valve, Basic pneumatic circuits, principle of working of power operated holding devices – chuck and clamping device, Pneumatic circuit design and analysis Illustrate the basic thermodynamics principles and different modes of

4 Understanding hydraulic and pneumatic circuits and systems Hydraulics: Introduction, basic concepts, Hydraulic fluids, Classification and properties of hydraulic fluids, Basic components, Fluid power generators, i.e. Gear, Vane, Piston pumps, linear and Rotary Actuators, Direction Control Valves, types, actuation methods, pressure control valves; pressure reducing valves, pressure relief valve, Flow control valves - simple and pressure compensated type. Circuit symbols, automatic cylinder reciprocating circuit & hydraulic circuit for robotic arm. Pneumatics: Introduction, Basic components, Source, storage and distribution, treatment of compressed air, linear and Rotary actuators, Direction control valves – types, actuation methods, pressure control valves, logic devices – twin pressure valve, shutter valve, time delay valve, Basic pneumatic circuits, principle of working of power operated holding devices – chuck and clamping device, Pneumatic circuit design and analysis Illustrate the basic thermodynamics principles and different modes of is an official topic in Module 3 of Fluid Mechanics & Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding hydraulic and pneumatic circuits and systems Hydraulics: Introduction, basic concepts, Hydraulic fluids, Classification and properties of hydraulic fluids, Basic components, Fluid power generators, i.e. Gear, Vane, Piston pumps, linear and Rotary Actuators, Direction Control Valves, types, actuation methods, pressure control valves; pressure reducing valves, pressure relief valve, Flow control valves - simple and pressure compensated type. Circuit symbols, automatic cylinder reciprocating circuit & hydraulic circuit for robotic arm. Pneumatics: Introduction, Basic components, Source, storage and distribution, treatment of compressed air, linear and Rotary actuators, Direction control valves – types, actuation methods, pressure control valves, logic devices – twin pressure valve, shutter valve, time delay valve, Basic pneumatic circuits, principle of working of power operated holding devices – chuck and clamping device, Pneumatic circuit design and analysis Illustrate the basic thermodynamics principles and different modes of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding hydraulic and pneumatic circuits and systems hydraulics: introduction, basic concepts, hydraulic fluids, classification and properties of hydraulic fluids, basic components, fluid power generators, i.e. gear, vane, piston pumps, linear and rotary actuators, direction control valves, types, actuation methods, pressure control valves; pressure reducing valves, pressure relief valve, flow control valves - simple and pressure compensated type. circuit symbols, automatic cylinder reciprocating circuit & hydraulic circuit for robotic arm. pneumatics: introduction, basic components, source, storage and distribution, treatment of compressed air, linear and rotary actuators, direction control valves – types, actuation methods, pressure control valves, logic devices – twin pressure valve, shutter valve, time delay valve, basic pneumatic circuits, principle of working of power operated holding devices – chuck and clamping device, pneumatic circuit design and analysis illustrate the basic thermodynamics principles and different modes of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding hydraulic and pneumatic circuits and systems Hydraulics: Introduction, basic concepts, Hydraulic fluids, Classification and properties of hydraulic fluids, Basic components, Fluid power generators, i.e. Gear, Vane, Piston pumps, linear and Rotary Actuators, Direction Control Valves, types, actuation methods, pressure control valves; pressure reducing valves, pressure relief valve, Flow control valves - simple and pressure compensated type. Circuit symbols, automatic cylinder reciprocating circuit & hydraulic circuit for robotic arm. Pneumatics:

Aspect	What to prepare
	Introduction, Basic components, Source, storage and distribution, treatment of compressed air, linear and Rotary actuators, Direction control valves – types, actuation methods, pressure control valves, logic devices – twin pressure valve, shutter valve, time delay valve, Basic pneumatic circuits, principle of working of power operated holding devices – chuck and clamping device, Pneumatic circuit design and analysis Illustrate the basic thermodynamics principles and different modes of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Understanding hydraulic and pneumatic circuits and systems Hydraulics: Introduction, basic concepts, Hydraulic fluids, Classification and properties of hydraulic fluids, Basic components, Fluid power generators, i.e. Gear, Vane, Piston pumps, linear and Rotary Actuators, Direction Control Valves, types, actuation methods, pressure control valves; pressure reducing valves, pressure relief valve, Flow control valves - simple and pressure compensated type. Circuit symbols, automatic cylinder reciprocating circuit & hydraulic circuit for robotic arm. Pneumatics: Introduction, Basic components, Source, storage and distribution, treatment of compressed air, linear and Rotary actuators, Direction control valves – types, actuation methods, pressure control valves, logic devices – twin pressure valve, shutter valve, time delay valve, Basic pneumatic circuits, principle of working of power operated holding devices – chuck and clamping device, Pneumatic circuit design and analysis Illustrate the basic thermodynamics principles and different modes of □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
systems

            Discuss basics of Hydraulics and Pneumatics
  M3.01                                                               1
Understanding
            and its applications
            Summarize the features and functions of
  M3.02     various components of hydraulic and                       4
Understanding
            pneumatic systems

            Identify various Hydraulic and Pneumatic
  M3.03                                                               2
Remembering
            symbols

            Describe the types of valves used in hydraulic
  M3.04                                                               4
Understanding
            and pneumatic systems
            Explain the working of different types of
  M3.05                                                               4
Understanding
            hydraulic and pneumatic circuits and systems

Contents:
Hydraulics: Introduction, basic concepts, Hydraulic fluids, Classification and
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

heat transfer

This module is presented from the official syllabus scope for Fluid Mechanics & Heat Transfer. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: heat transfer
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain basic concepts in thermodynamics 3 Understanding Interpret the concepts of conduction to solve

Explain basic concepts in thermodynamics 3 Understanding Interpret the concepts of conduction to solve is an official topic in Module 4 of Fluid Mechanics & Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain basic concepts in thermodynamics 3 Understanding Interpret the concepts of conduction to solve using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain basic concepts in thermodynamics 3 understanding interpret the concepts of conduction to solve should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain basic concepts in thermodynamics 3 Understanding Interpret the concepts of conduction to solve means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain basic concepts in thermodynamics 3 Understanding Interpret the concepts of conduction to solve □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

problems involving steady state heat 6 Applying conduction in plane walls.

problems involving steady state heat 6 Applying conduction in plane walls. is an official topic in Module 4 of Fluid Mechanics & Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify problems involving steady state heat 6 Applying conduction in plane walls. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, problems involving steady state heat 6 applying conduction in plane walls. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What problems involving steady state heat 6 Applying conduction in plane walls. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-□□ problems involving steady state heat 6 Applying conduction in plane walls. □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Illustrate concepts of convection and radiation 5 Applying Introduction to Thermodynamics : Role of Thermodynamics in Engineering and Science, Types of Systems, Thermodynamic Equilibrium, Properties, State, Process and Cycle, Elementary introduction to Zeroth, First and Second laws of thermodynamics, Heat and Work Interactions for various non-flow and flow processes Heat Transfer: Conduction, convection and radiation-Fourier's law of thermal conduction. Thermal conductivity, conduction through a plane wall and composite plane wall – Numerical problems. Thermal radiation – reflection, absorption and transmission, absorptivity, reflectivity and transmissivity, Stefan - Boltzman's law of total radiation. Newton Rikhman equation of Thermal convection, free convection, forced convection. Simple numerical problems on convection and radiation. Text/ Reference: T/R Book Title/Author Rattan S S, "Fluid Mechanics & Hydraulic Machines", Khanna Publishing House, T1 New Delhi, 2019 Yunus. A Cengel and Afshin. J Ghajar, "Heat and Mass Transfer: Fundamentals T2 and Applications", McGraw-Hill Education Nag P K, "Engineering Thermodynamics", McGraw-Hill Education, Sixth Edition T3 2017 Ramamritham. S, "Hydraulics, Fluid Mechanics And Fluid Machines", R1 DhanpatRai&Sons,Delhi, 2006. P. N Modi and S. M. Seth, "Hydraulics and Fluid Mechanics Including Hydraulics R2 Machines", 22nd Edition, Standard Book House, 2019 R3 Anthony Esposito,"Fluid Power with applications",Pearson Education. Werner Deppert and Kurt Stoll,"Pneumatic Control-An introduction to the R4 principles", Vogel-Verlag. Rajput R K, "A Textbook of Heat and Mass Transfer" S. Chand Publications, R5 2019 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/112104118/ 2 https://freevideolectures.com/course/89/fluid-mechanics 3 http://www.efunda.com/formulae/heat_transfer/home/overview.cfm 4 http://www.nptelvideos.in/2012/12/heat-and-mass-transfer.html 5 http://web.mit.edu/hml/ncfmf.html

Illustrate concepts of convection and radiation 5 Applying Introduction to Thermodynamics : Role of Thermodynamics in Engineering and Science, Types of Systems, Thermodynamic Equilibrium, Properties, State, Process and Cycle, Elementary introduction to Zeroth, First and Second laws of thermodynamics, Heat and Work Interactions for various non-flow and flow processes Heat Transfer: Conduction, convection and radiation-Fourier's law of thermal conduction. Thermal conductivity, conduction through a plane wall and composite plane wall – Numerical problems. Thermal radiation – reflection, absorption and transmission, absorptivity, reflectivity and transmissivity, Stefan - Boltzman's law of total radiation. Newton Rikhman equation of Thermal convection, free convection, forced convection. Simple numerical problems on convection and radiation. Text/ Reference: T/R Book Title/Author Rattan S S, "Fluid

Mechanics & Hydraulic Machines”, Khanna Publishing House, T1 New Delhi, 2019 Yunus. A Cengel and Afshin. J Ghajar, “Heat and Mass Transfer: Fundamentals T2 and Applications”, McGraw-Hill Education Nag P K, “Engineering Thermodynamics”, McGraw-Hill Education, Sixth Edition T3 2017 Ramamritham. S, “Hydraulics, Fluid Mechanics And Fluid Machines”, R1 DhanpatRai&Sons,Delhi, 2006. P. N Modi and S. M. Seth, “Hydraulics and Fluid Mechanics Including Hydraulics R2 Machines”, 22nd Edition, Standard Book House, 2019 R3 Anthony Esposito,“Fluid Power with applications”,Pearson Education. Werner Deppert and Kurt Stoll,“Pneumatic Control-An introduction to the R4 principles”, Vogel-Verlag. Rajput R K, “A Textbook of Heat and Mass Transfer” S. Chand Publications, R5 2019 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/112104118/ 2 https://freevideolectures.com/course/89/fluid-mechanics 3 http://www.efunda.com/formulae/heat_transfer/home/overview.cfm 4 http://www.nptelvideos.in/2012/12/heat-and-mass-transfer.html 5 http://web.mit.edu/hml/ncfmf.html is an official topic in Module 4 of Fluid Mechanics & Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate concepts of convection and radiation 5 Applying Introduction to Thermodynamics : Role of Thermodynamics in Engineering and Science, Types of Systems, Thermodynamic Equilibrium, Properties, State, Process and Cycle, Elementary introduction to Zeroth, First and Second laws of thermodynamics, Heat and Work Interactions for various non-flow and flow processes Heat Transfer: Conduction, convection and radiation-Fourier's law of thermal conduction. Thermal conductivity, conduction through a plane wall and composite plane wall – Numerical problems. Thermal radiation – reflection, absorption and transmission, absorptivity, reflectivity and transmissivity, Stefan - Boltzman's law of total radiation. Newton Rikhman equation of Thermal convection, free convection, forced convection. Simple numerical problems on convection and radiation. Text/ Reference: T/R Book Title/Author Rattan S S, “Fluid Mechanics & Hydraulic Machines”, Khanna Publishing House, T1 New Delhi, 2019 Yunus. A Cengel and Afshin. J Ghajar, “Heat and Mass Transfer: Fundamentals T2 and Applications”, McGraw-Hill Education Nag P K, “Engineering Thermodynamics”, McGraw-Hill Education, Sixth Edition T3 2017 Ramamritham. S, “Hydraulics, Fluid

Mechanics And Fluid Machines", R1 DhanpatRai&Sons,Delhi, 2006. P. N Modi and S. M. Seth, "Hydraulics and Fluid Mechanics Including Hydraulics R2 Machines", 22nd Edition, Standard Book House, 2019 R3 Anthony Esposito,"Fluid Power with applications",Pearson Education. Werner Deppert and Kurt Stoll,"Pneumatic Control-An introduction to the R4 principles", Vogel-Verlag. Rajput R K, "A Textbook of Heat and Mass Transfer" S. Chand Publications, R5 2019 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/112104118/ 2 https://freevideolectures.com/course/89/fluid-mechanics 3 http://www.efunda.com/formulae/heat_transfer/home/overview.cfm 4 http://www.nptelvideos.in/2012/12/heat-and-mass-transfer.html 5 http://web.mit.edu/hml/ncfmf.html using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate concepts of convection and radiation 5 applying introduction to thermodynamics : role of thermodynamics in engineering and science, types of systems, thermodynamic equilibrium, properties, state, process and cycle, elementary introduction to zeroth, first and second laws of thermodynamics, heat and work interactions for various non-flow and flow processes heat transfer: conduction, convection and radiation-fourier's law of thermal conduction. thermal conductivity, conduction through a plane wall and composite plane wall – numerical problems. thermal radiation – reflection, absorption and transmission, absorptivity, reflectivity and transmissivity, stefan - boltzman's law of total radiation. newton rikhman equation of thermal convection, free convection, forced convection. simple numerical problems on convection and radiation. text/ reference: t/r book title/author rattan s s, "fluid mechanics & hydraulic machines", khanna publishing house, t1 new delhi, 2019 yunus. a cengel and afshin. j ghajar, "heat and mass transfer: fundamentals t2 and applications", mcgraw-hill education nag p k, "engineering thermodynamics", mcgraw-hill education, sixth edition t3 2017 ramamritham. s, "hydraulics, fluid mechanics and fluid machines", r1 dhanpatrai&sons,delhi, 2006. p. n modi and s. m. seth, "hydraulics and fluid mechanics including hydraulics r2 machines", 22nd edition, standard book house, 2019 r3 anthony esposito,"fluid power with applications",pearson education. werner deppert and kurt stoll,"pneumatic control-an introduction to the r4 principles", vogel-verlag. rajput r k, "a textbook of heat and mass transfer" s. chand publications, r5 2019 online resources: sl.no website link 1 https://nptel.ac.in/courses/112104118/ 2 https://freevideolectures.com/course/89/fluid-mechanics 3 http://www.efunda.com/formulae/heat_transfer/home/overview.cfm 4 http://www.nptelvideos.in/2012/12/heat-and-mass-transfer.html 5 http://web.mit.edu/hml/ncfmf.html should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate concepts of convection and radiation 5 Applying Introduction to Thermodynamics : Role of Thermodynamics in Engineering and Science, Types of Systems, Thermodynamic Equilibrium, Properties, State, Process and Cycle, Elementary introduction to Zeroth, First and Second laws of thermodynamics, Heat and Work Interactions for various non-flow and flow processes Heat Transfer: Conduction, convection and radiation-Fourier's law of thermal conduction. Thermal conductivity, conduction through a plane wall and composite plane wall – Numerical problems. Thermal radiation – reflection, absorption and transmission, absorptivity, reflectivity and transmissivity, Stefan - Boltzman's law of total radiation. Newton Rikhman equation of Thermal convection, free convection, forced convection. Simple numerical problems on convection and radiation. Text/ Reference: T/R Book Title/Author Rattan S S, "Fluid Mechanics & Hydraulic Machines", Khanna Publishing House, T1 New Delhi, 2019 Yunus. A Cengel and Afshin. J Ghajar, "Heat and Mass Transfer: Fundamentals T2 and Applications", McGraw-Hill Education Nag P K, "Engineering Thermodynamics", McGraw-Hill Education, Sixth Edition T3 2017 Ramamritham. S, "Hydraulics, Fluid Mechanics And Fluid Machines", R1 DhanpatRai&Sons,Delhi, 2006. P. N Modi and S. M. Seth, "Hydraulics and Fluid Mechanics Including Hydraulics R2 Machines", 22nd Edition, Standard Book House, 2019 R3 Anthony Esposito,"Fluid Power with applications",Pearson Education. Werner Deppert and Kurt Stoll,"Pneumatic Control-An introduction to the R4 principles", Vogel-Verlag. Rajput R K, "A Textbook of Heat and Mass Transfer" S. Chand Publications, R5 2019 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/112104118/ 2 https://freevideolectures.com/course/89/fluid-mechanics 3 http://www.efunda.com/formulae/heat_transfer/home/overview.cfm 4 http://www.nptelvideos.in/2012/12/heat-and-mass-transfer.html 5 http://web.mit.edu/hml/ncfmf.html means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Illustrate concepts of convection and radiation 5 Applying Introduction to Thermodynamics : Role of Thermodynamics in Engineering and Science, Types of Systems, Thermodynamic Equilibrium, Properties, State, Process and Cycle, Elementary introduction to Zeroth, First and Second laws of thermodynamics, Heat and Work Interactions for various non-flow and flow processes Heat Transfer: Conduction, convection and radiation-Fourier's law of thermal conduction. Thermal conductivity, conduction through a plane wall and composite plane wall – Numerical problems. Thermal radiation – reflection, absorption and transmission, absorptivity, reflectivity and transmissivity, Stefan - Boltzman's law of total radiation. Newton Rikhman equation of Thermal convection, free convection, forced convection. Simple numerical problems on convection and radiation. Text/ Reference: T/R Book Title/Author Rattan S S, “Fluid Mechanics & Hydraulic Machines”, Khanna Publishing House, T1 New Delhi, 2019 Yunus. A Cengel and Afshin. J Ghajar, “Heat and Mass Transfer: Fundamentals T2 and Applications”, McGraw-Hill Education Nag P K, “Engineering Thermodynamics”, McGraw-Hill Education, Sixth Edition T3 2017 Ramamritham. S, “Hydraulics, Fluid Mechanics And Fluid Machines”, R1 DhanpatRai&Sons,Delhi, 2006. P. N Modi and S. M. Seth, “Hydraulics and Fluid Mechanics Including Hydraulics R2 Machines”, 22nd Edition, Standard Book House, 2019 R3 Anthony Esposito,“Fluid Power with applications”,Pearson Education. Werner Deppert and Kurt Stoll,“Pneumatic Control-An introduction to the R4 principles”, Vogel-Verlag. Rajput R K, “A Textbook of Heat and Mass Transfer” S. Chand Publications, R5 2019 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/112104118/ 2 https://freevideolectures.com/course/89/fluid-mechanics 3 http://www.efunda.com/formulae/heat_transfer/home/overview.cfm 4 http://www.nptelvideos.in/2012/12/heat-and-mass-transfer.html 5 http://web.mit.edu/hml/ncfmf.html □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
heat transfer
  M4.01     Explain basic concepts in thermodynamics            3
Understanding
            Interpret the concepts of conduction to solve
  M4.02     problems involving steady state heat                6
Applying
            conduction in plane walls.
  M4.03     Illustrate concepts of convection and radiation     5
Applying

            Series Test II                                      1

Contents:
Introduction to Thermodynamics : Role of Thermodynamics in Engineering and
Science,
Types of Systems, Thermodynamic Equilibrium, Properties, State, Process and
Cycle,
Elementary introduction to Zeroth, First and Second laws of thermodynamics, Heat
and
Work Interactions for various non-flow and flow processes
Heat Transfer: Conduction, convection and radiation-Fourier's law of thermal
conduction.
Thermal conductivity, conduction through a plane wall and composite plane wall –
Numerical problems. Thermal radiation – reflection, absorption and transmission,
absorptivity, reflectivity and transmissivity, Stefan - Boltzman's law of total
radiation.
Newton Rikhman equation of Thermal convection, free convection, forced
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Explain fluid properties 3 Understanding Explain the Fluid Pressure and the methods of. (3 marks)

Answer: Begin with the standard definition or identity of *Explain fluid properties 3 Understanding Explain the Fluid Pressure and the methods of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding measurement Illustrate the principle of working of various. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding measurement Illustrate the principle of working of various*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain pressure measuring devices and solve simple 4 Applying problems on those Interpret the terms total pressure & buoyancy. (3 marks)

Answer: Begin with the standard definition or identity of *pressure measuring devices and solve simple 4 Applying problems on those Interpret the terms total pressure & buoyancy*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain and calculate the total pressure on immersed 4 Applying surfaces Properties of fluid: Density, Specific gravity, Specific Weight, Specific Volume, Dynamic Viscosity, Kinematic Viscosity, Surface tension, Capillarity, Vapour Pressure, Compressibility. Numerical problems on fluid properties. Fluid Pressure & Pressure Measurement: Fluid pressure, Pressure head, Pressure intensity, Concept of vacuum and gauge pressures, atmospheric pressure, absolute pressure, Simple and differential manometers, Bourdons pressure gauge, Concept of Total pressure on immersed bodies, center of pressure, Simple problems on pressure measurements Buoyancy and flotation, Archimedes' principle, stability of immersed and floating bodies,

determination of metacentric height. Problems on total pressure. Investigate the applications of laws of conservation of momentum, mass. (3 marks)

Answer: Begin with the standard definition or identity of *and calculate the total pressure on immersed 4 Applying surfaces Properties of fluid: Density, Specific gravity, Specific Weight, Specific Volume, Dynamic Viscosity, Kinematic Viscosity, Surface tension, Capillarity, Vapour Pressure, Compressibility. Numerical problems on fluid properties. Fluid Pressure & Pressure Measurement: Fluid pressure, Pressure head, Pressure intensity, Concept of vacuum and gauge pressures, atmospheric pressure, absolute pressure, Simple and differential manometers, Bourdons pressure gauge, Concept of Total pressure on immersed bodies, center of pressure, Simple problems on pressure measurements Buoyancy and flotation, Archimedes' principle, stability of immersed and floating bodies, determination of metacentric height. Problems on total pressure. Investigate the applications of laws of conservation of momentum, mass*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain Describe the types of fluid flows 2 Understanding Apply the momentum and energy equations to. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the types of fluid flows 2 Understanding Apply the momentum and energy equations to*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain fluid flow problems based on an analysis of 4 Applying the various system specifications Illustrate the concept of flow through pipes,. (3 marks)

Answer: Begin with the standard definition or identity of *fluid flow problems based on an analysis of 4 Applying the various system specifications Illustrate the concept of flow through pipes,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Applying orifices & notches List the losses of head in pipes and identify. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying orifices & notches List the losses of head in pipes and identify*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Major losses and Minor losses and solve 4 Applying problems Fluid Flow: Types of fluid flows, Path line and Stream line, Continuity equation, Bernoulli's theorem, Principle of operation of Venturimeter, Orifice meter and Pitot tube, Derivations for discharge, coefficient of discharge, numerical problems. Flow Through Pipes, Orifices & Notches: Darcy's equation and Chezy's equation for frictional losses (No derivation), Minor losses in pipes, Numerical problems to estimate major and minor losses. Orifices: types of orifices, Vena contracta, coefficient of contraction, coefficient of velocity, coefficient of discharge, simple problems. Notches: Types –Rectangular notches , triangular notch , trapezoidal notch , discharge over notches – equations only, simple numerical problems Distinguish the components and basic circuits in hydraulic and pneumatic. (3 marks)

Answer: Begin with the standard definition or identity of *Major losses and Minor losses and solve 4 Applying problems Fluid Flow: Types of fluid flows, Path line and Stream line, Continuity equation, Bernoulli's theorem, Principle of operation of Venturimeter, Orifice meter and Pitot tube, Derivations for discharge, coefficient of discharge, numerical problems. Flow Through Pipes, Orifices & Notches: Darcy's equation and Chezy's* *equation for frictional losses (No derivation), Minor losses in pipes, Numerical problems to estimate major and minor losses. Orifices: types of orifices, Vena contracta, coefficient of contraction, coefficient of velocity, coefficient of discharge, simple problems. Notches: Types –Rectangular notches , triangular notch , trapezoidal notch , discharge over notches – equations only, simple numerical problems Distinguish the components and basic circuits in hydraulic and pneumatic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

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Module 3

Define or explain 1 Understanding and its applications Summarize the features and functions of. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding and its applications Summarize the features and functions of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain various components of hydraulic and 4 Understanding pneumatic systems Identify various Hydraulic and Pneumatic. (3 marks)

Answer: Begin with the standard definition or identity of *various components of hydraulic and 4 Understanding pneumatic systems Identify various Hydraulic and Pneumatic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 2 Remembering symbols Describe the types of valves used in hydraulic. (3 marks)

Answer: Begin with the standard definition or identity of *2 Remembering symbols Describe the types of valves used in hydraulic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 4 Understanding and pneumatic systems Explain the working of different types of. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding and pneumatic systems Explain the working of different types of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 4

Define or explain Explain basic concepts in thermodynamics 3 Understanding Interpret the concepts of conduction to solve. (3 marks)

Answer: Begin with the standard definition or identity of *Explain basic concepts in thermodynamics 3 Understanding Interpret the concepts of conduction to solve*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain problems involving steady state heat 6 Applying conduction in plane walls.. (3 marks)

Answer: Begin with the standard definition or identity of *problems involving steady state heat 6 Applying conduction in plane walls.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate concepts of convection and radiation 5 Applying Introduction to Thermodynamics : Role of Thermodynamics in Engineering and Science, Types of Systems, Thermodynamic Equilibrium, Properties, State, Process and Cycle, Elementary introduction to Zeroth, First and Second laws of thermodynamics, Heat and Work Interactions for various non-flow and flow processes Heat Transfer: Conduction, convection and radiation-Fourier's law of thermal conduction. Thermal conductivity, conduction through a plane wall and composite plane wall – Numerical problems. Thermal radiation – reflection, absorption and transmission, absorptivity, reflectivity and transmissivity, Stefan - Boltzman's law of total radiation. Newton Rikhman equation of Thermal convection, free convection, forced convection. Simple numerical problems on convection and radiation. Text/ Reference: T/R Book Title/Author Rattan S S, “Fluid Mechanics & Hydraulic Machines”, Khanna Publishing House, T1 New Delhi, 2019 Yunus. A Cengel and Afshin. J Ghajar, “Heat and Mass Transfer: Fundamentals T2 and Applications”, McGraw-Hill Education Nag P K, “Engineering Thermodynamics”, McGraw-Hill Education, Sixth Edition T3 2017 Ramamritham. S, “Hydraulics, Fluid Mechanics And Fluid Machines”, R1 DhanpatRai&Sons,Delhi, 2006. P. N Modi and S. M. Seth, “Hydraulics and Fluid Mechanics Including Hydraulics R2 Machines”, 22nd Edition, Standard Book House, 2019 R3 Anthony Esposito,“Fluid Power with applications”,Pearson Education. Werner Deppert and Kurt Stoll,“Pneumatic Control-An introduction to the R4 principles”, Vogel-Verlag. Rajput R K, “A Textbook of Heat and Mass Transfer” S. Chand Publications, R5 2019 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/112104118/ 2 https://freevideolectures.com/course/89/fluid-mechanics 3 http://www.efunda.com/formulae/heat_transfer/home/overview.cfm 4 http://www.nptelvideos.in/2012/12/heat-and-mass-transfer.html 5 http://web.mit.edu/hml/ncfmf.html. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate concepts of convection and radiation 5 Applying Introduction to Thermodynamics : Role of Thermodynamics in Engineering and Science, Types of Systems, Thermodynamic Equilibrium, Properties, State, Process and Cycle, Elementary introduction to Zeroth, First and Second laws of thermodynamics, Heat and Work Interactions for various non-flow and flow processes Heat Transfer: Conduction, convection and radiation-Fourier's law of thermal conduction. Thermal conductivity, conduction through a plane wall and composite plane wall – Numerical problems. Thermal radiation – reflection, absorption and transmission, absorptivity, reflectivity and transmissivity, Stefan - Boltzman's law of total radiation. Newton Rikhman equation of Thermal convection, free convection, forced convection. Simple numerical problems on convection and radiation. Text/ Reference: T/R Book Title/Author Rattan S S, "Fluid Mechanics & Hydraulic Machines", Khanna Publishing House, T1 New Delhi, 2019 Yunus. A Cengel and Afshin. J Ghajar, "Heat and Mass Transfer: Fundamentals T2 and Applications", McGraw-Hill Education Nag P K, "Engineering Thermodynamics", McGraw-Hill Education, Sixth Edition T3 2017 Ramamritham. S, "Hydraulics, Fluid Mechanics And Fluid Machines", R1 DhanpatRai&Sons,Delhi, 2006. P. N Modi and S. M. Seth, "Hydraulics and Fluid Mechanics Including Hydraulics R2 Machines", 22nd Edition, Standard Book House, 2019 R3 Anthony Esposito,"Fluid Power with applications",Pearson Education. Werner Deppert and Kurt Stoll,"Pneumatic Control-An introduction to the R4 principles", Vogel-Verlag. Rajput R K, "A Textbook of Heat and Mass Transfer" S. Chand Publications, R5 2019 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/112104118/ 2 https://freevideolectures.com/course/89/fluid-mechanics 3 http://www.efunda.com/formulae/heat_transfer/home/overview.cfm 4 http://www.nptelvideos.in/2012/12/heat-and-mass-transfer.html 5 http://web.mit.edu/hml/ncfmf.html*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain fluid properties 3 Understanding Explain the Fluid Pressure and the methods of**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Describe the types of fluid flows 2 Understanding Apply the momentum and energy equations to**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **1 Understanding and its applications Summarize the features and functions of**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain basic concepts in thermodynamics 3 Understanding Interpret the concepts of conduction to solve**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link https://nptel.ac.in/courses/112104118/
- https://freevideolectures.com/course/89/fluid-mechanics
- http://www.efunda.com/formulae/heat_transfer/home/overview.cfm
- http://www.nptelvideos.in/2012/12/heat-and-mass-transfer.html
  http://web.mit.edu/hml/ncfmf.html

IN-PAGE SEARCH

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